Analytical Chemistry Cumulative Exam
March 30, 2007
(Chemical Equilibria and Titrations)

1. (30 points) A **50.00 mL** solution of **0.100 M H_2CO_3** is titrated with **0.100 M NaOH**. Sketch out the titration curve of pH vs. volume of 0.100 M NaOH added. The curve should follow the titration **from 0.00 mL of NaOH to 150.00 mL of NaOH solution**. Try to be quantitative in drawing your titration curve. Label your y and x axes. Indicate where the equivalence point(s) should appear and calculate the pH of at least two points along the curve. (Note: K_{a1}=4.45 x 10^{-7}, and K_{a2} = 4.69 x 10^{-11})

2. Using UV-Vis absorbance to follow titrations
Draw the titration curve in terms of relative absorbance vs. volume of titrant, for each of the scenarios a, b, and c below. Show your work. No credit will be given for a graph without the supporting data and explanation. Assume that the total volume is not significantly changed during the titration (so that you don't have to account for dilution effects.)

Here are the relevant conditions:
• The reaction of interest is: A + T → P
• Your analyte is "A" and is the only species that absorbs visible radiation in the starting sample solution. "A" does not absorb at 700 nm, but does absorb at 400 nm.
• You titrate your sample solution with the titrant 'T", which absorbs at 700 nm, but does not absorb at 400 nm.
• A product is formed, "P", from the reaction of "A" with "T". "P" has the same absorption characteristics as "A", in that it does not absorb at 700 nm, but does absorb at 400 nm. In addition, it has the same extinction coefficient as "A".
• When one molecule of "A" reacts with one molecule of "T", the reaction is essentially complete and irreversible…a good condition of a titration.

(a) (10 points) Draw the titration curve of absorbance at 400 nm as a function of titrant added. (Also, indicate the endpoint on the titration curve.)
(b) (10 points) Draw the titration curve of absorbance at 700 nm as a function of titrant added. (Also, indicate the endpoint on the titration curve.)
(c) (10 points) What are the advantages of titrations over direct UV-Visible absorbance spectroscopy of a sample of unknown analyte concentration?

3. Dilute NaI is introduced into a solution that is 0.060 M in $AgNO_3$ and 0.070 M in $Bi(NO_3)_3$. The K_{sp} for AgI is 8.3 x 10^{-17} and the K_{sp} for BiI_3 is 8.1 x 10^{-19}.
(a) (10 points) Which iodide precipitates first? What I- concentration is needed to initiate the precipitation of the first iodide?
(b) (10 points) What is the concentration of the cation forming the *less* soluble iodide when the more soluble iodide begins to form?

4. (20 points) A sample contains the amino acid alanine, **CH_3CHNH_2COOH**, (MW = **89.1 g/mol**), plus inert matter. A **3.50 g** sample is digested and analyzed by the Kjeldahl method. The ammonia produced is distilled and collected in **50.0 mL** of **0.150 M H_2SO_4**, and **7.50 mL** of **0.100 M NaOH** is required for back titration. What is the percent alanine in the sample? (Assume *all* acidic protons of the left-over H_2SO_4 are titrated by the NaOH.)